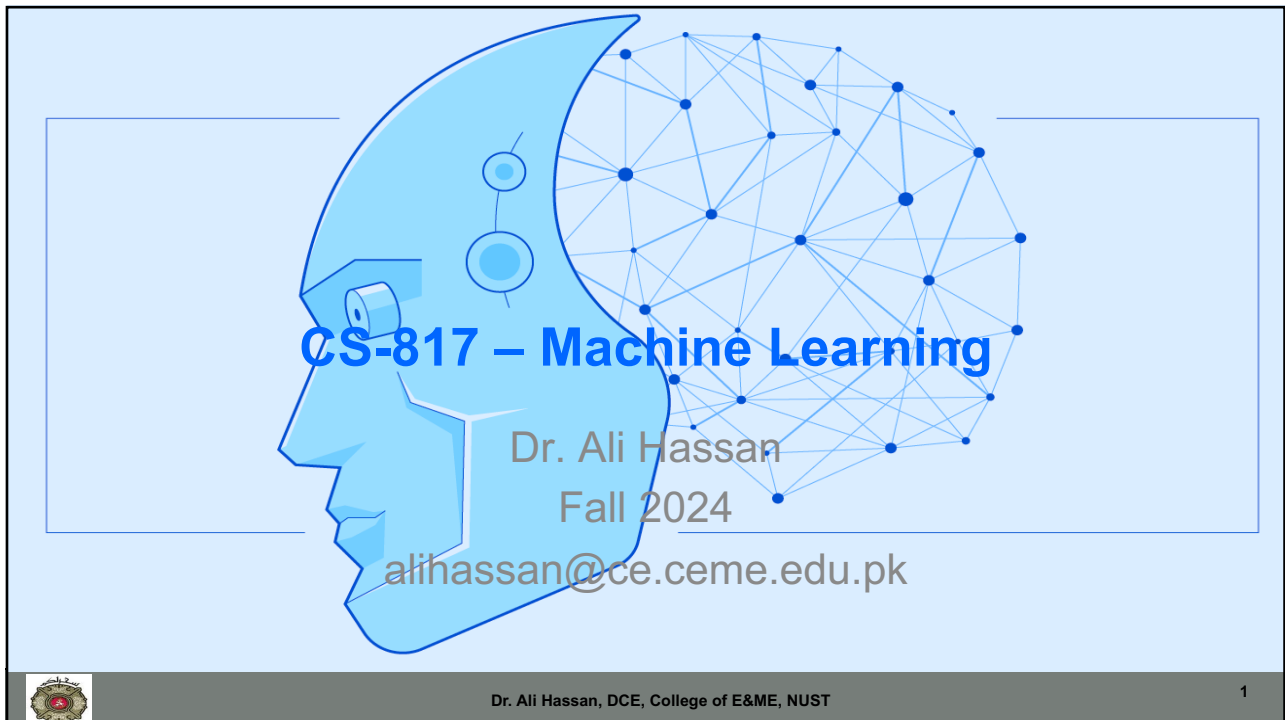


A blue graphic of a human head profile facing left, with a stylized brain composed of a network of nodes and lines. The text "CS-817 – Machine Learning" is written in blue across the head. Below it, "Dr. Ali Hassan" and "Fall 2024" are written in brown. The email "alihassan@ce.ceme.edu.pk" is at the bottom in brown. A small NUST logo is in the bottom left corner.

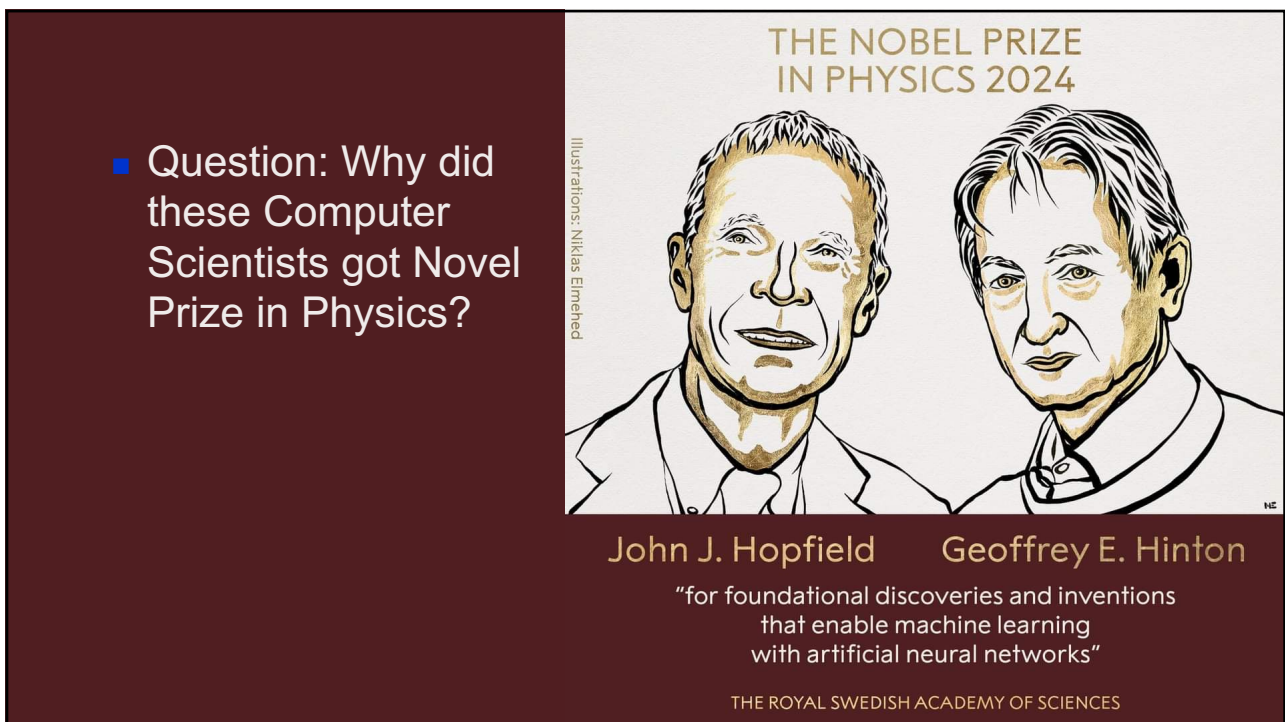
CS-817 – Machine Learning

Dr. Ali Hassan
Fall 2024
alihassan@ce.ceme.edu.pk

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A dark red slide with a white box on the left containing a question. On the right, there is a graphic for the Nobel Prize in Physics 2024 featuring portraits of John J. Hopfield and Geoffrey E. Hinton. The text "THE NOBEL PRIZE IN PHYSICS 2024" is at the top right. Below the portraits are their names and the award citation. The NUST logo is in the bottom left corner.

■ Question: Why did these Computer Scientists get Nobel Prize in Physics?

THE NOBEL PRIZE
IN PHYSICS 2024

Illustrations: Niklas Elmehed

John J. Hopfield Geoffrey E. Hinton

"for foundational discoveries and inventions that enable machine learning with artificial neural networks"

THE ROYAL SWEDISH ACADEMY OF SCIENCES

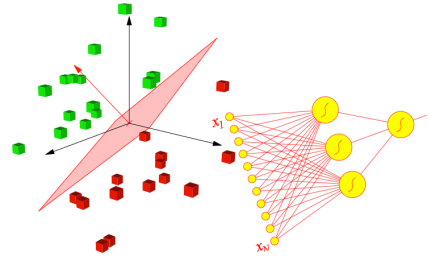
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Announcements

- Assignment 2
 - Questions
- Makeup Class
 - Date?



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Previous Lecture

- Logistic Regression
- Regularization



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Outline

- Feature Selection
 - Filter Based Methods
 - Wrapper Based Methods
 - PCA



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FEATURE SELECTION



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Why Feature Selection

- The curse of Dimensionality



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Feature Selection

What constitutes a good set of features for classification?

- Select features having high discrimination power while ignoring or paying less attention to the rest.
- Given a set of **n** features, the role of **feature selection** is to select a subset of size **k** ($k < n$) that leads to the smallest classification error.



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Why is feature selection important?

- Eliminates **redundant** and **irrelevant** features.
- Classifier requires **less** training examples for training (i.e., curse of dimensionality).
- **Faster** and more **accurate** classification.



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Feature Selection Methods

- Feature selection is an **optimization** problem.
 - Search the space of possible feature subsets.
 - Pick the one that is optimal or near-optimal with respect to a certain criterion.

Search strategies

- Exhaustive
- Heuristic
- Randomized

Evaluation strategies

- Filter methods
- Wrapper methods



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Exhaustive Search

- Assuming m features, an exhaustive search would require:
 - Examining all $\binom{n}{k}$ possible subsets of size k .
 - Selecting the subset that performs the best according to the **evaluation** function.
- The number of subsets grows combinatorially, making exhaustive search impractical.
- Iterative procedures are often used but they **cannot** guarantee the selection of the optimal subset.



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Naïve Search

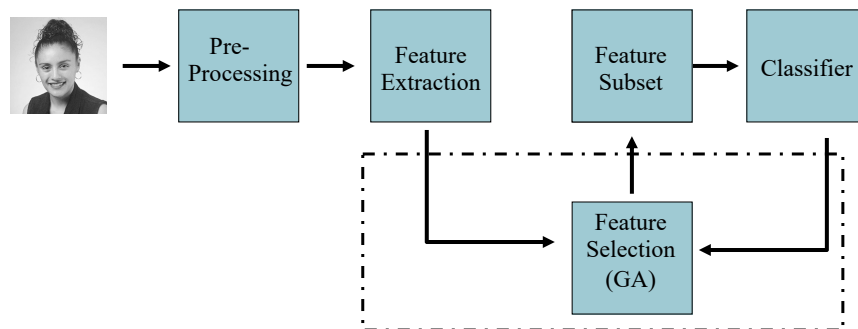
- Sort the given n features in order of their probability of correct recognition.
- Select the top k features from this sorted list.
- Disadvantage
 - Feature correlation is not considered.
 - Best pair of features may not even contain the best individual feature.



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Feature Selection using Genetic Algorithms (GAs) (randomized search)

GAs provide a simple, general, and powerful framework for feature selection.



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Evaluation Strategies

- Filter Methods
 - Evaluation is independent of the classification algorithm or its error criteria.
- Wrapper Methods
 - Evaluation uses a criterion related to the classification algorithm.
- Wrapper methods provide more accurate solutions than filter methods, but in general are more computationally expensive.



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Sequential forward selection (SFS) (heuristic search)

- First, the **best single feature** is selected (i.e., using some criterion function).
- Then, **pairs of features** are formed using one of the remaining features and this best feature, and the best pair is selected.
- Next, **triplets of features** are formed using one of the remaining features and these two best features, and the best triplet is selected.
- This procedure continues until evaluation criterion **increases monotonically** or a predefined number of features are selected.



SFS performs best when the optimal subset is **small**.



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SFS Algorithm

Algorithm 5.1 Pseudo Code for Sequential Forwarding Selection (SFS)

Input: $F = \{f_i | i = 1 \dots D\}$

Output: S^d where $d < D$

```

1:  $S^0 \leftarrow \{\}$ 
2: oldcriterion  $\leftarrow 0$ 
3: newcriterion  $\leftarrow 1$ 
4: while newcriterion > oldcriterion do
5:   oldcriterion  $\leftarrow J(S^d)$ 
6:    $f^+ := \operatorname{argmax}_{f_i \notin S^d} J(S^d + f_i)$ 
7:    $S^{d+1} \leftarrow \{S^d \cup f^+\}$ 
8:   newcriterion  $\leftarrow J(S^{d+1})$ 
9:    $d \leftarrow d + 1$ 
10: end while
  
```

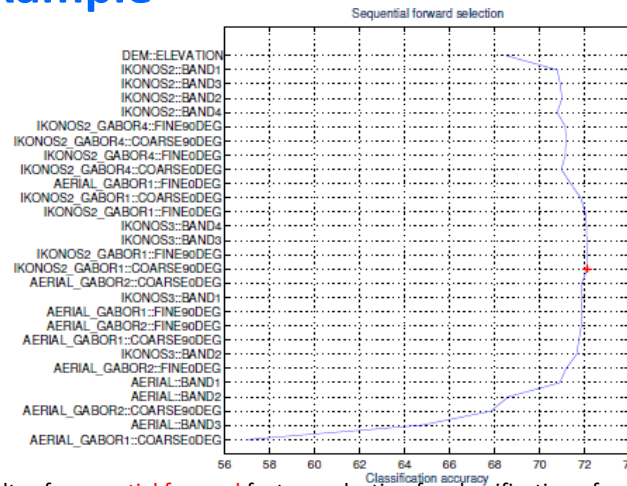


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Example



Results of **sequential forward** feature selection for classification of a satellite image using 28 features. x-axis shows the classification accuracy (%) and y-axis shows the features added at each iteration (the first iteration is at the bottom). The highest accuracy value is shown with a star.



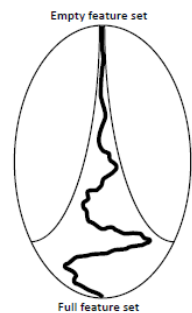
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Sequential backward selection (SBS) (heuristic search)

- First, the criterion function is computed for all d features.
- Then, each **feature is deleted one at a time**, the criterion function is computed for all subsets with $d - 1$ features, and the **worst feature is discarded**.
- Next, each feature among the remaining $d - 1$ is deleted one at a time, and the worst feature is discarded to form a subset with $d - 2$ features.
- This procedure continues until a predefined number of features are left.



SBS performs best when the optimal subset is **large**.



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Sequential floating selection (SFFS and SFBS)

- An extension to LRS with flexible backtracking capabilities
 - Rather than fixing the values of L and R, floating methods determine these values from the data.
 - The dimensionality of the subset during the search can be thought to be “floating” up and down
- There are two floating methods:
 - Sequential floating forward selection (SFFS)
 - Sequential floating backward selection (SFBS)

P. Pudil, J. Novovicova, J. Kittler, Floating search methods in feature selection, Pattern Recognition Lett. 15 (1994) 1119–1125.



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Sequential floating selection (SFFS and SFBS)

- SFFS
 - Sequential floating forward selection (SFFS) starts from the empty set.
 - After each forward step, SFFS performs backward steps as long as the objective function increases.
- SFBS
 - Sequential floating backward selection (SFBS) starts from the full set.
 - After each backward step, SFBS performs forward steps as long as the objective function increases.



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SFFS

Algorithm 5.1 Pseudo Code for Sequential Forwarding Selection (SFS)

Input: $F = \{f_i | i = 1 \dots D\}$

Output: S^d where $d < D$

```

1:  $S^0 \leftarrow \{\}$ 
2: oldcriterion  $\leftarrow 0$ 
3: newcriterion  $\leftarrow 1$ 
4: while newcriterion > oldcriterion do
5:   oldcriterion  $\leftarrow J(S^d)$ 
6:    $f^+ := \operatorname{argmax}_{f_i \notin S} J(S^d + f_i)$ 
7:    $S^{d+1} \leftarrow \{S^d \cup f^+\}$ 
8:   newcriterion  $\leftarrow J(S^{d+1})$ 
9:    $d \leftarrow d + 1$ 
10: end while

```



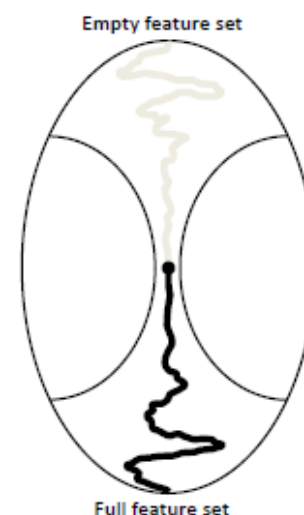
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Bidirectional Search (BDS)

- BDS applies SFS and SBS simultaneously:
 - SFS is performed from the empty set
 - SBS is performed from the full set
- To guarantee that SFS and SBS converge to the same solution
 - Features already selected by SFS are not removed by SBS
 - Features already removed by SBS are not selected by SFS



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DIMENSIONALITY REDUCTION



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Motivation

- Doing Exhaustive search is very expensive – **non-feasible**
- Doing Wrapper based feature selection (SFS,SBS, SFFS, etc) again very **expensive**
- Doing filter based feature selection is **suboptimal**
- Solution??? Try to **automate** this method



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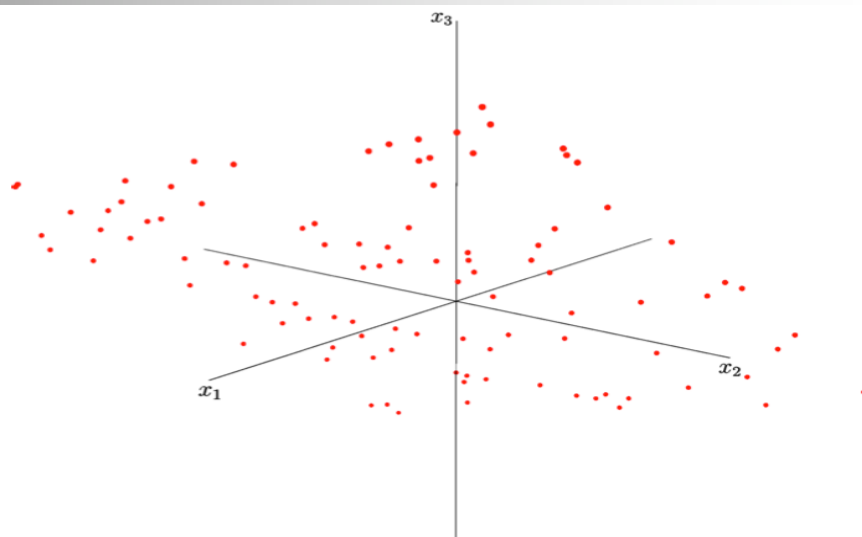
Spread of Data

- Often data varies in only some limited directions
- Can't spot low dimensional data by looking at numbers

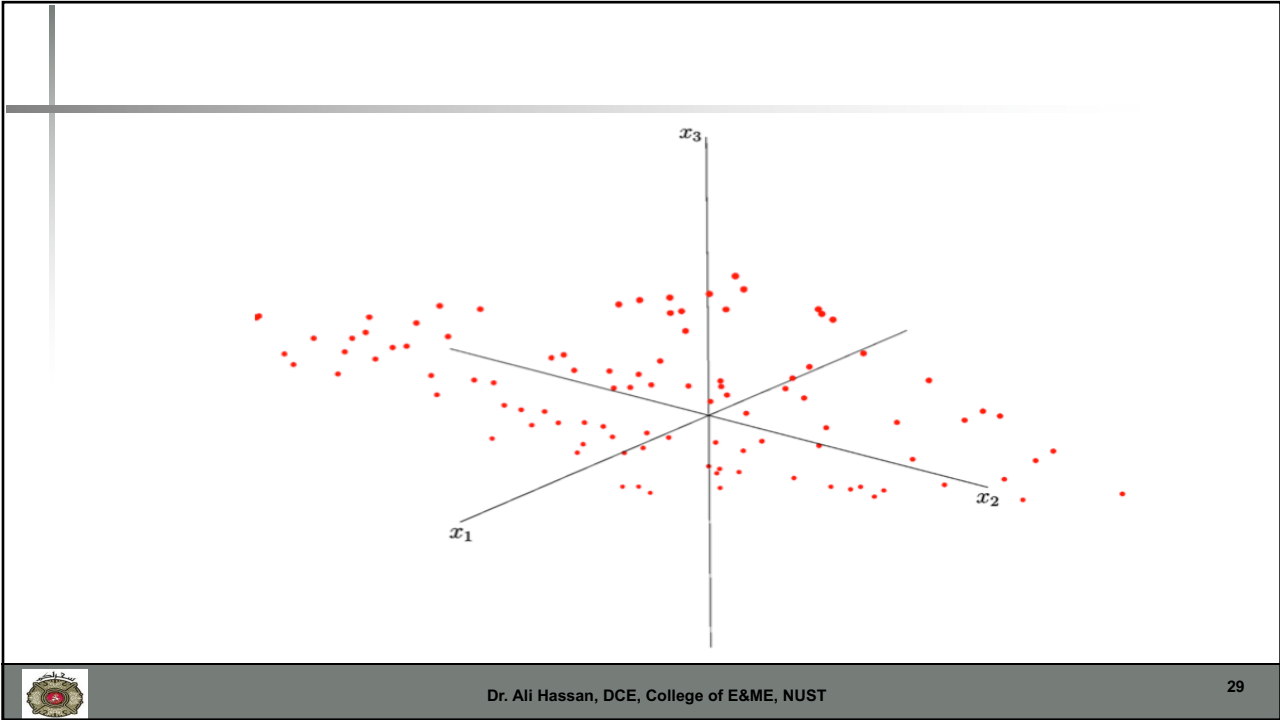


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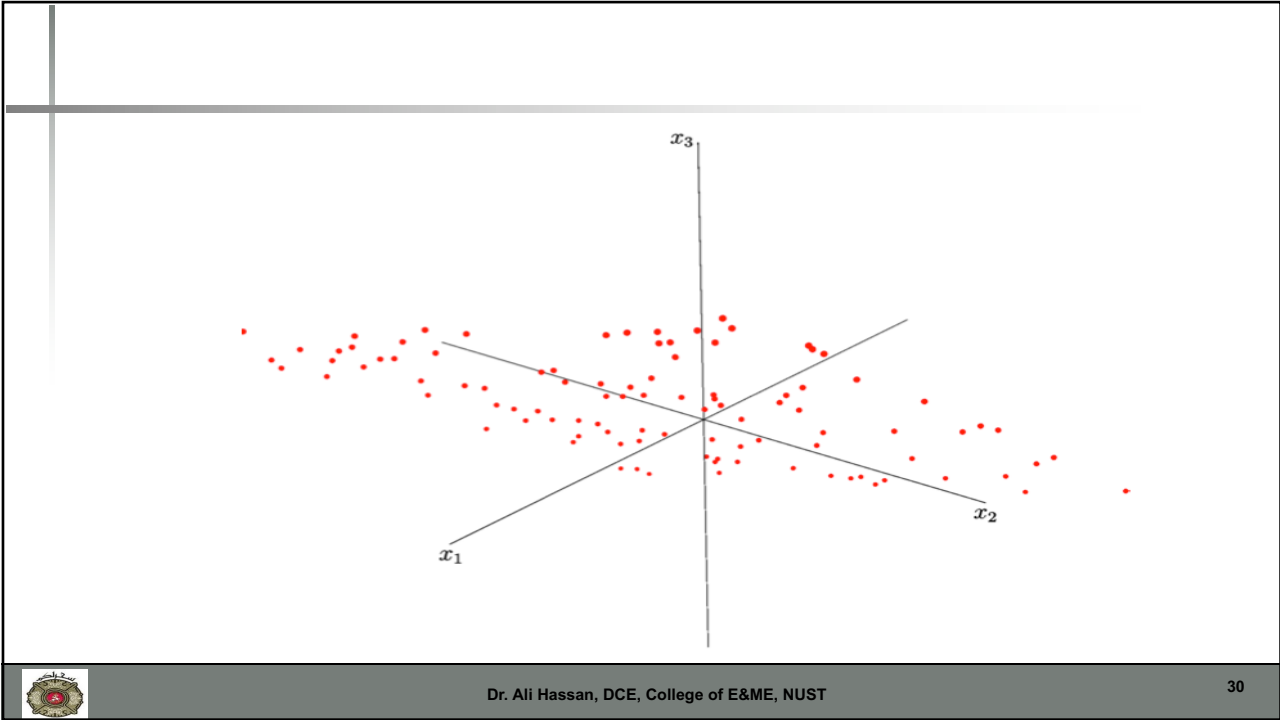
Example



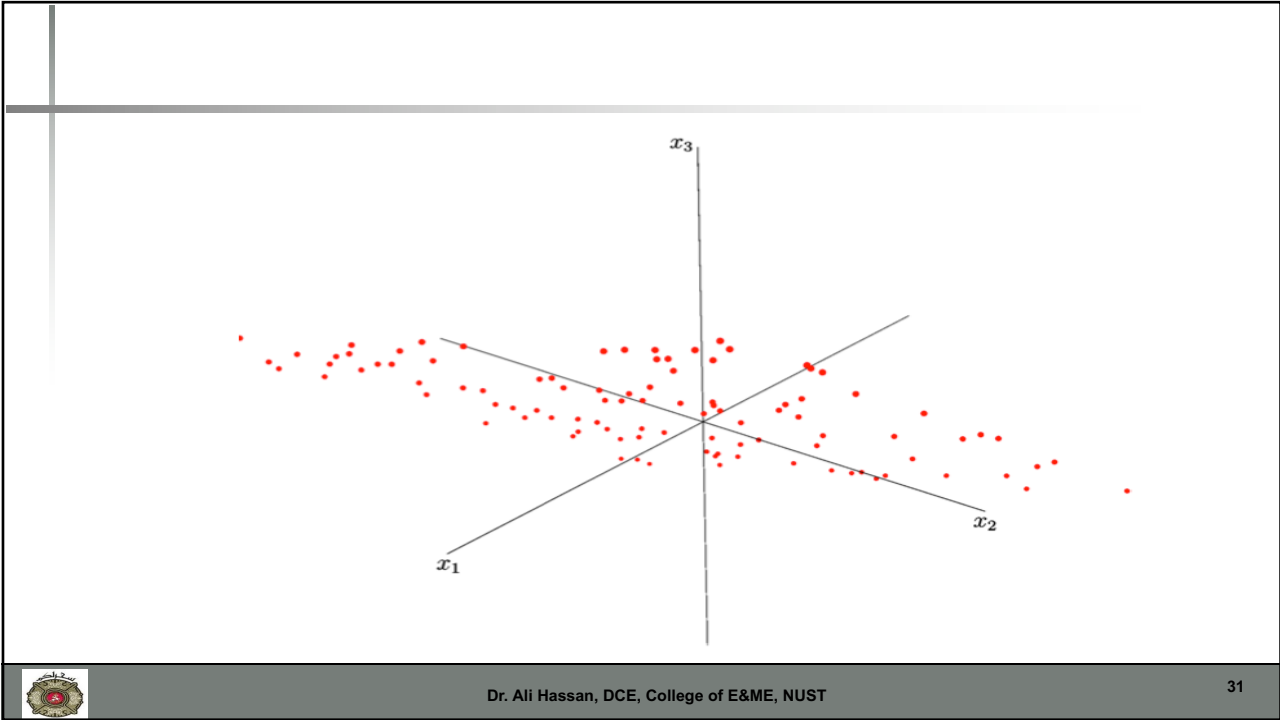
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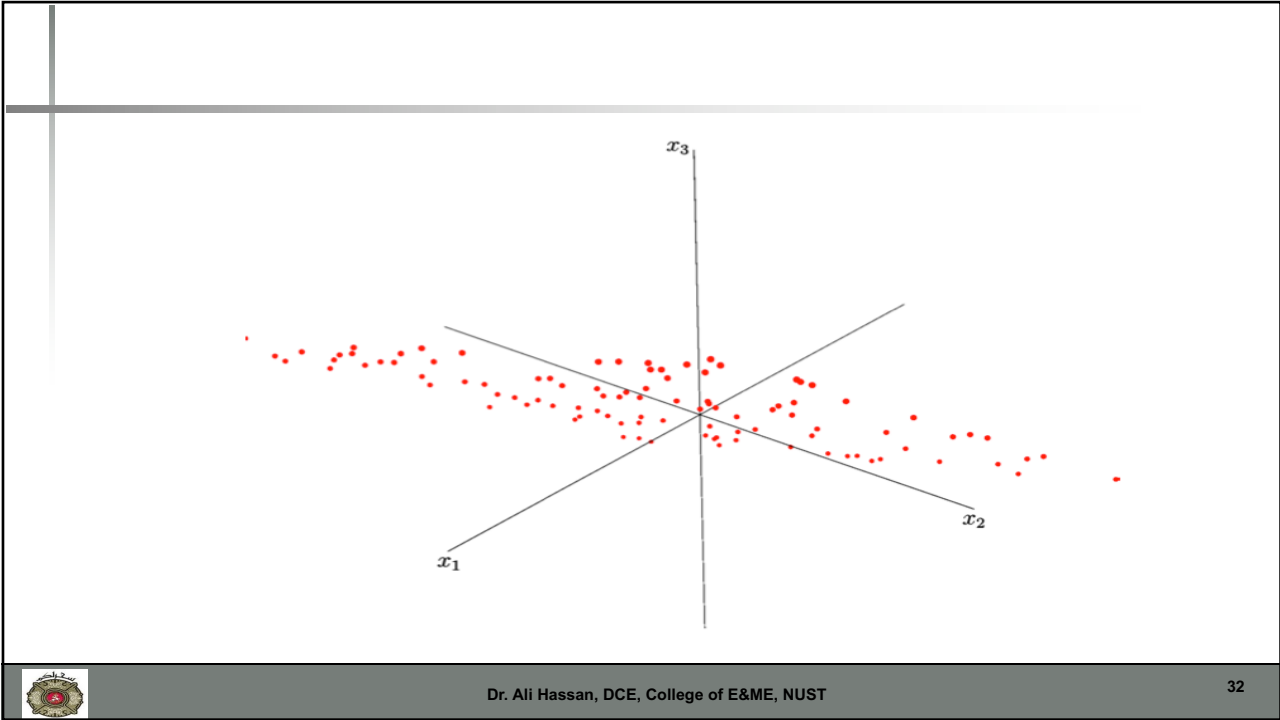
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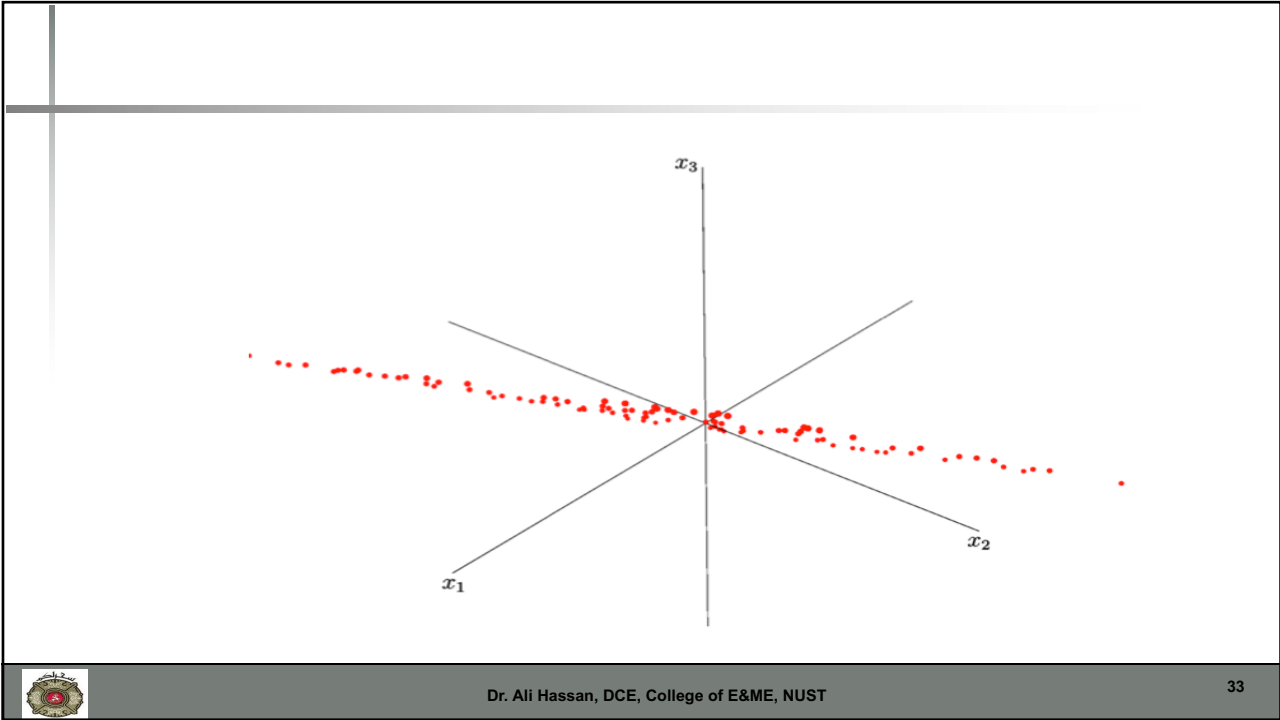
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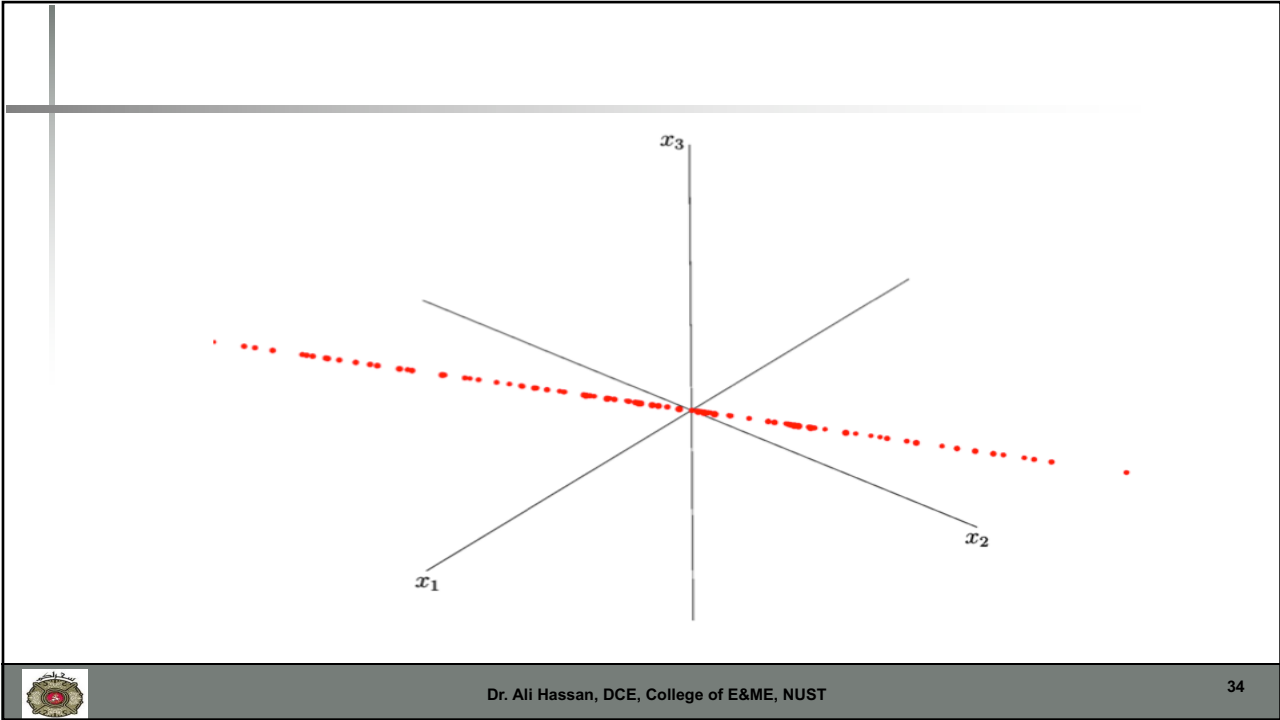
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Dimensionality Reduction

- Reduce dimensions by projecting onto low dimensional subspace with maximum variation
- You can consider this as dropping non-necessary axis and rotating the remaining axis



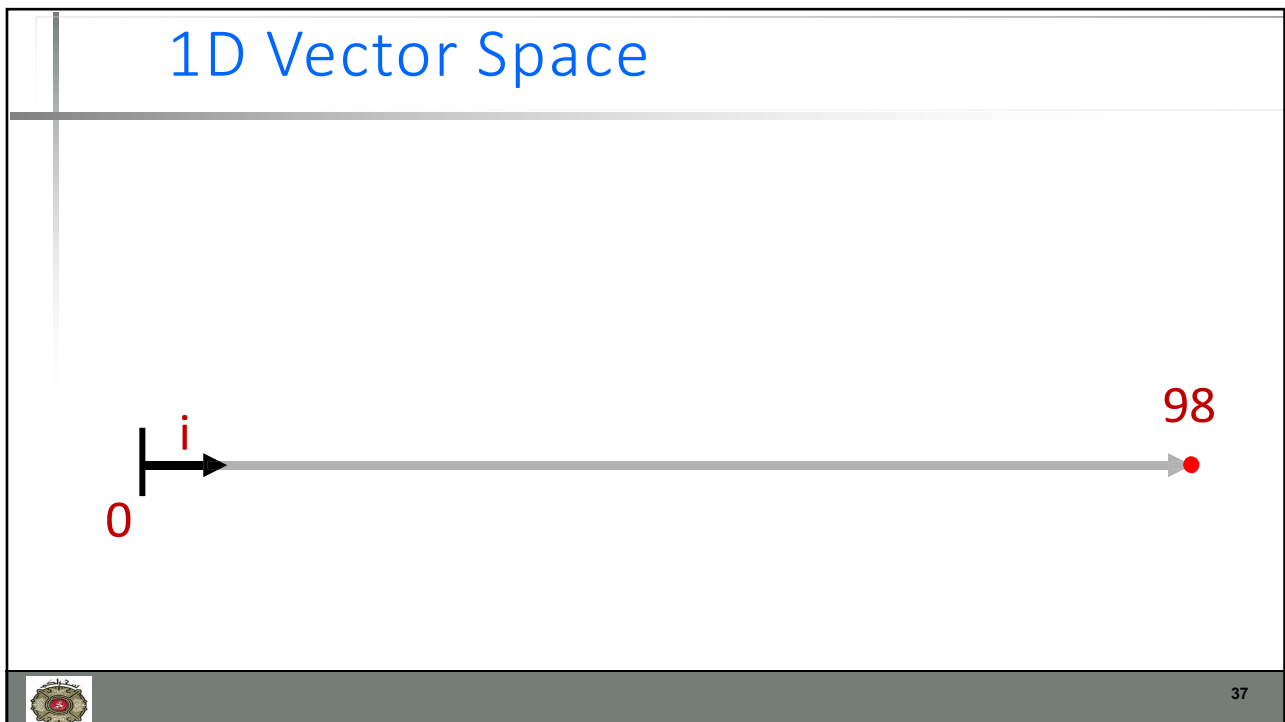
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Data Representation as Vectors

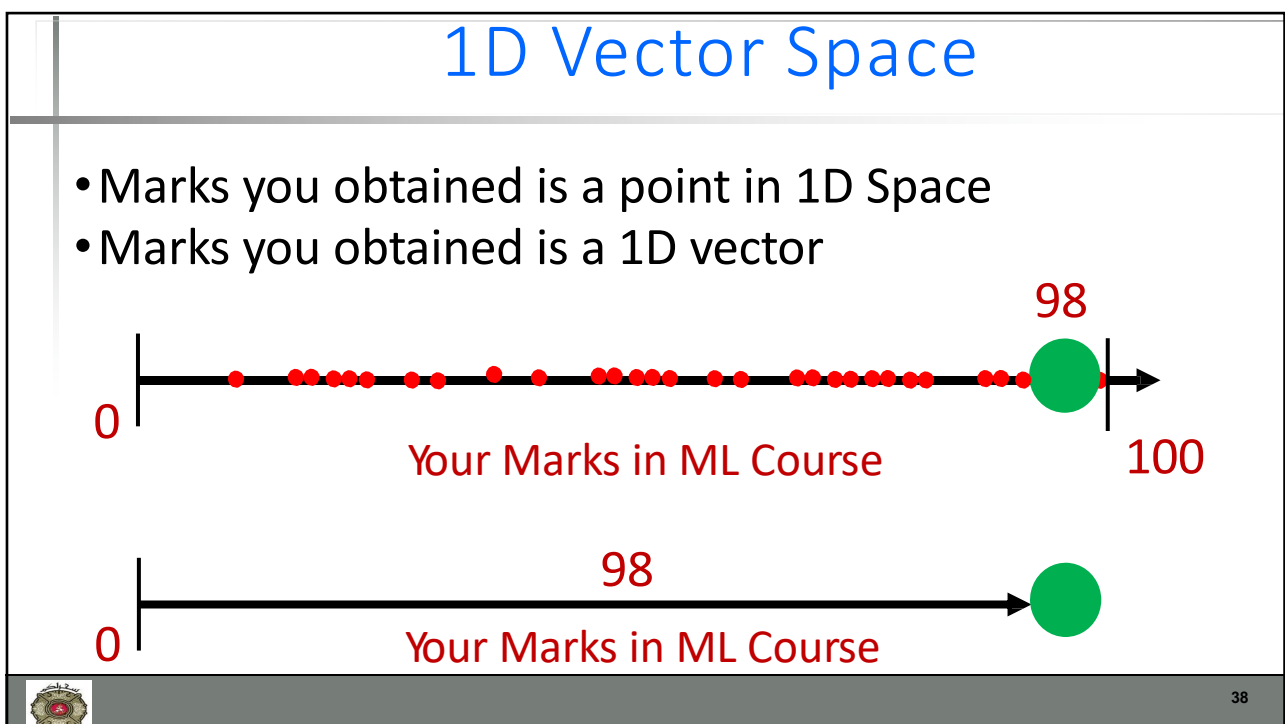
- Data can also be represented as Vectors and Points in a Vector Space
- We shall be studying
 - Vector Spaces
 - Vector Subspaces
 - Dimension
 - Basis
 - Spanning
 - Eigen Values



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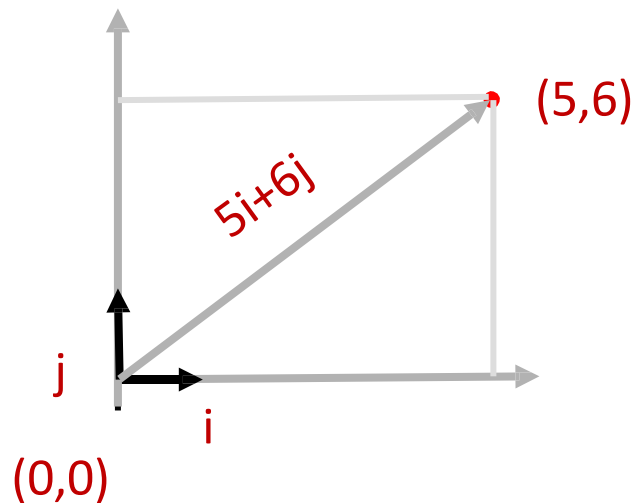


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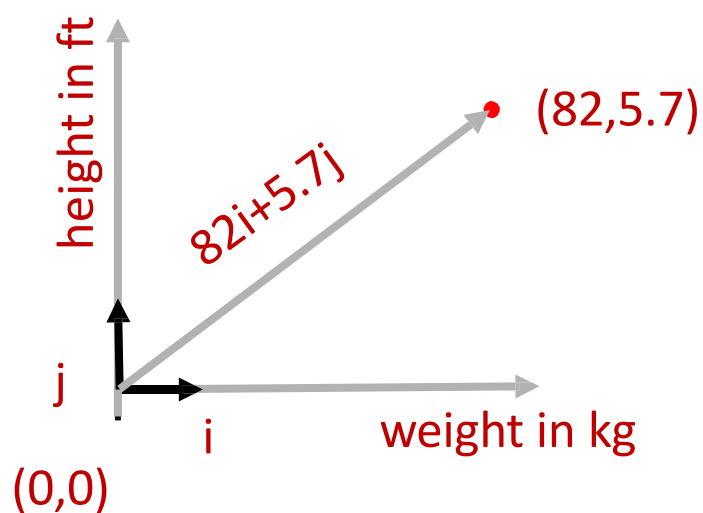
2D Vector Space



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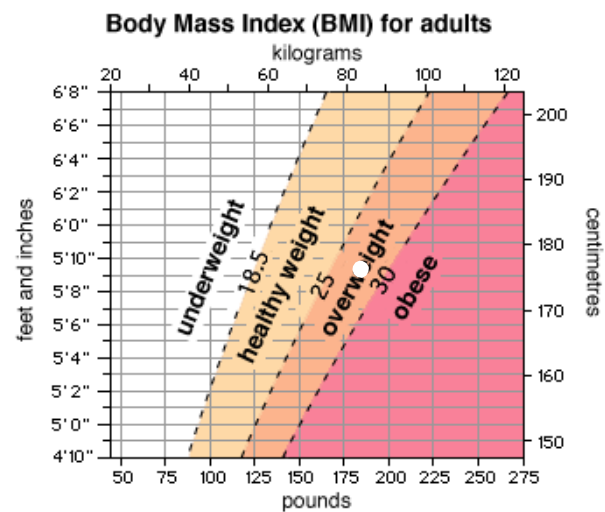
2D Vector Space Example



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Example of 2-D Vector Space

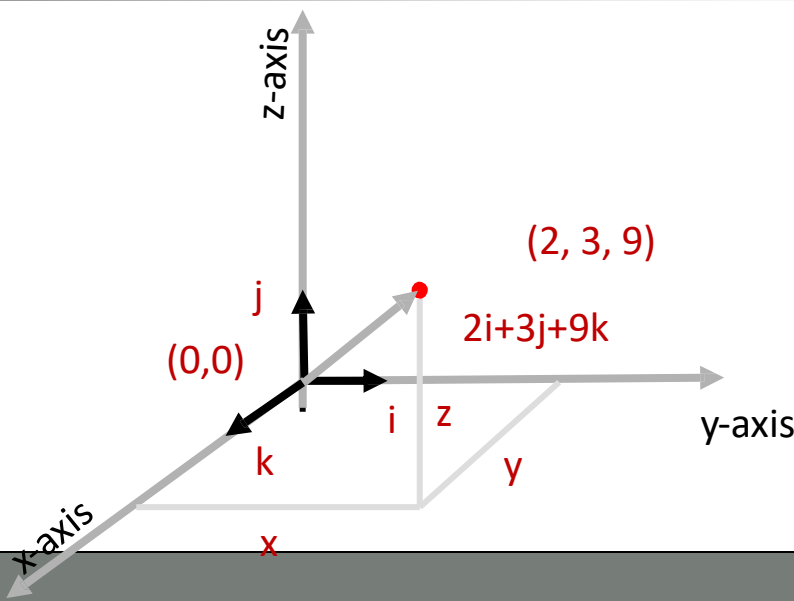


Source: National Institutes of Health/National Heart, Lung, and Blood Institute
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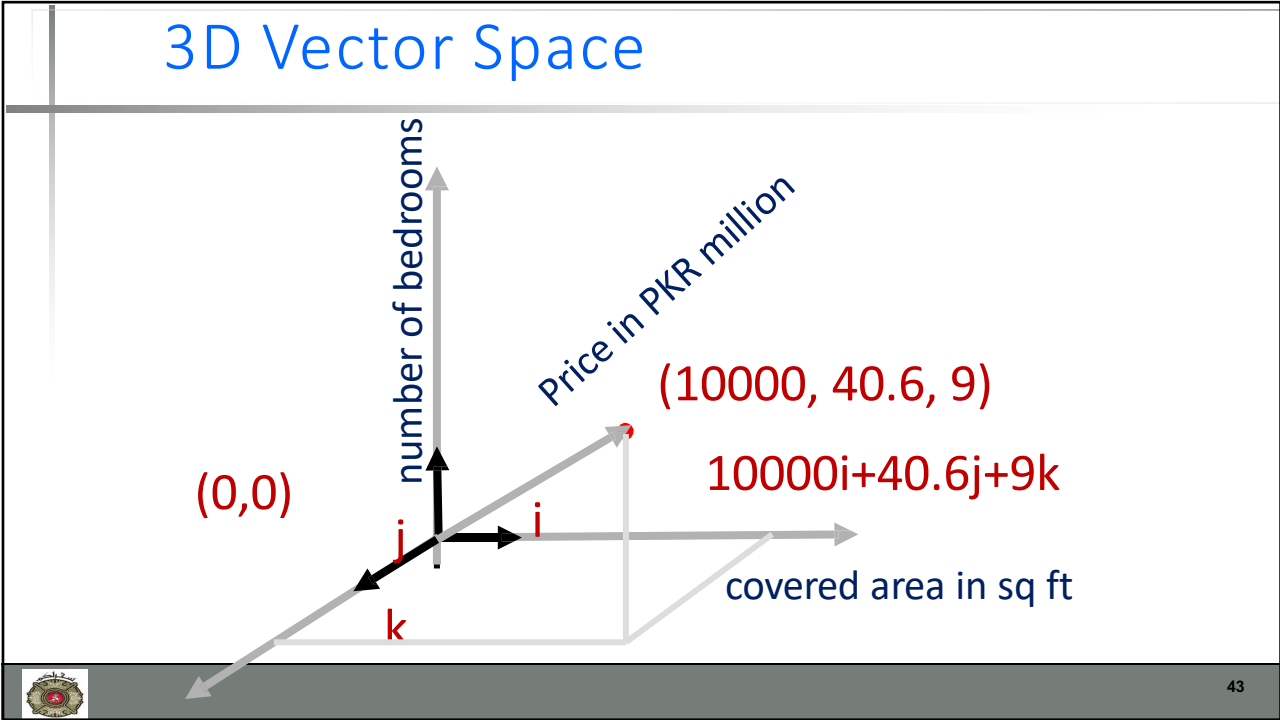
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3D Vector Space

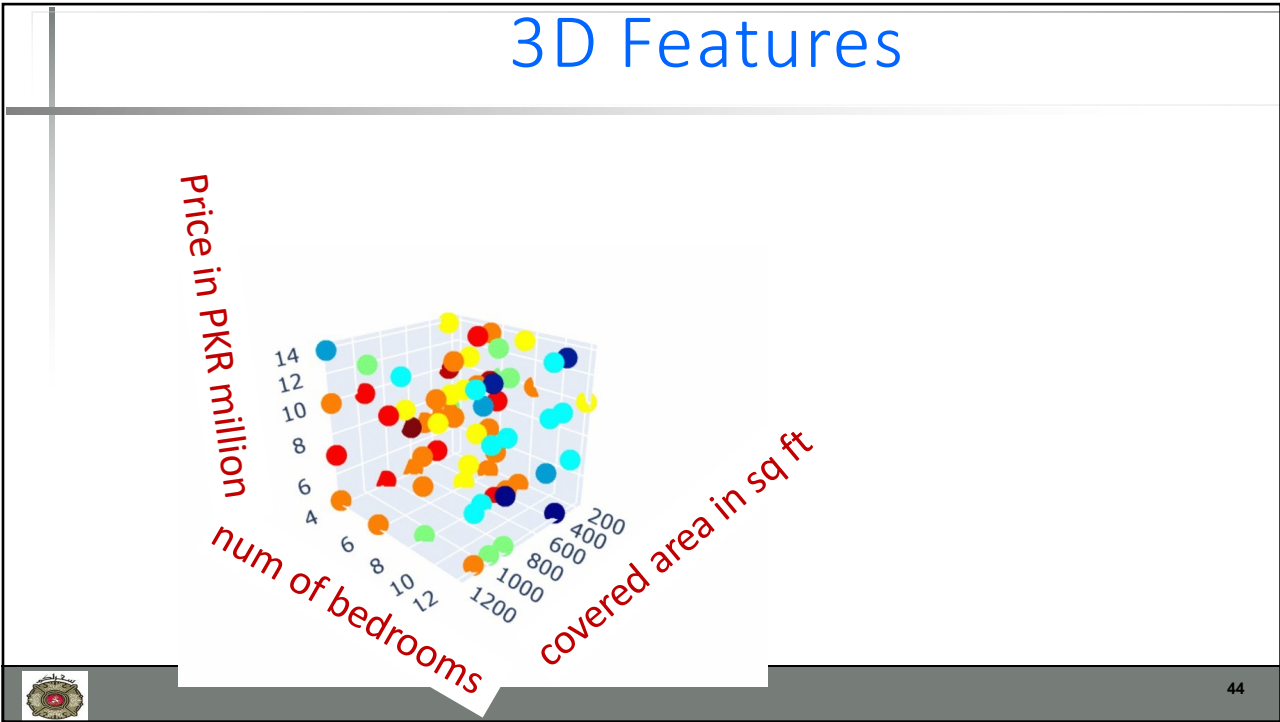


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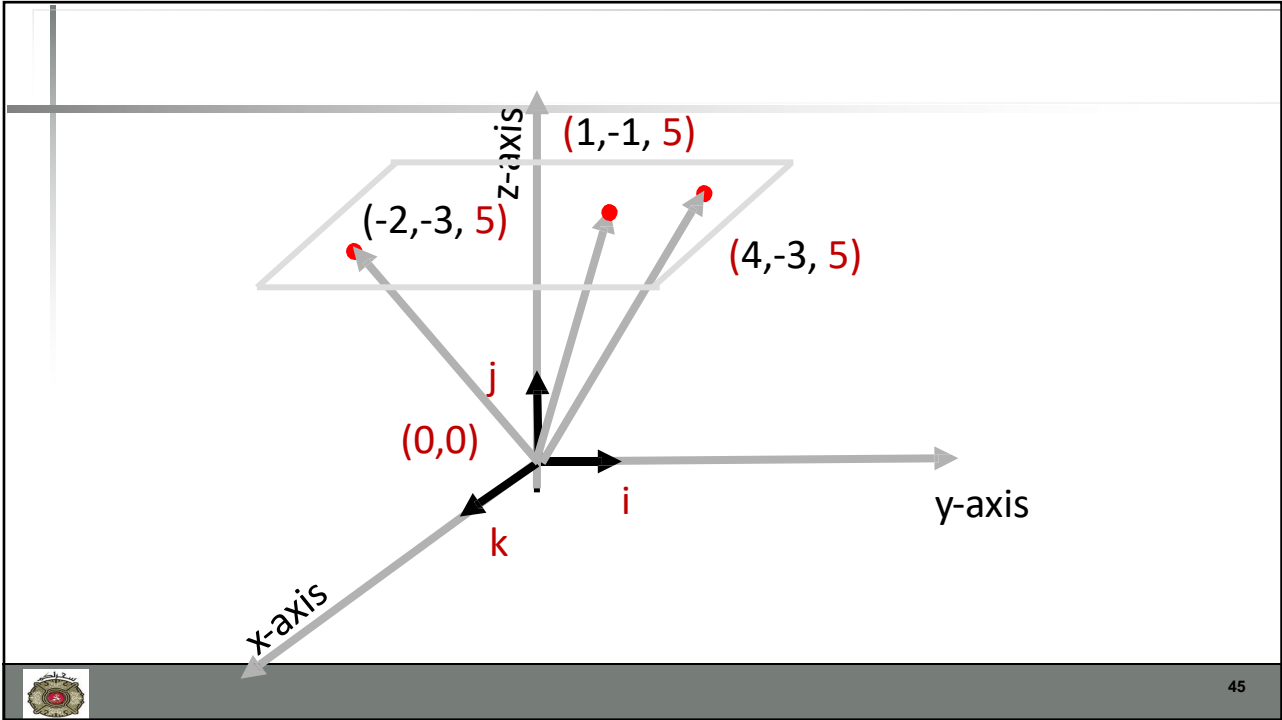
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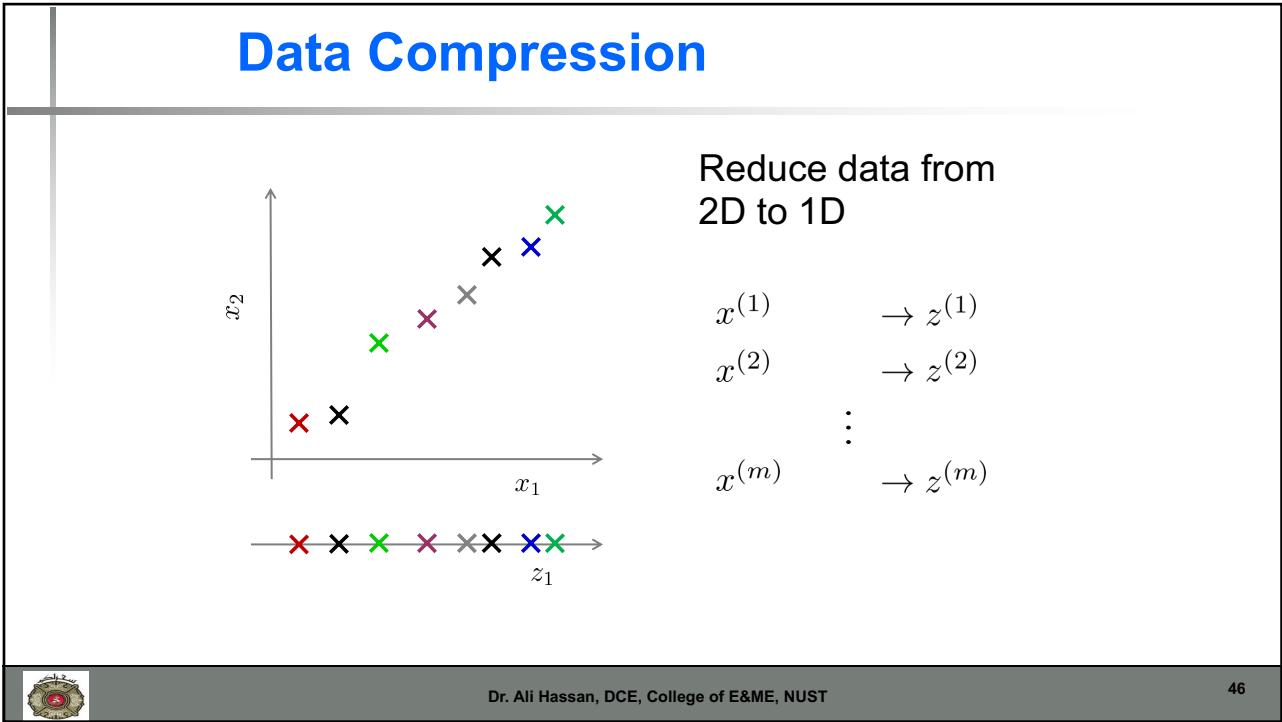
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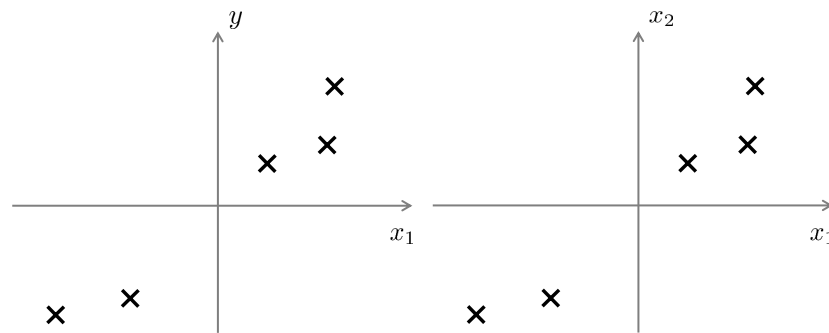


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PCA is not linear regression



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Principal Component Analysis

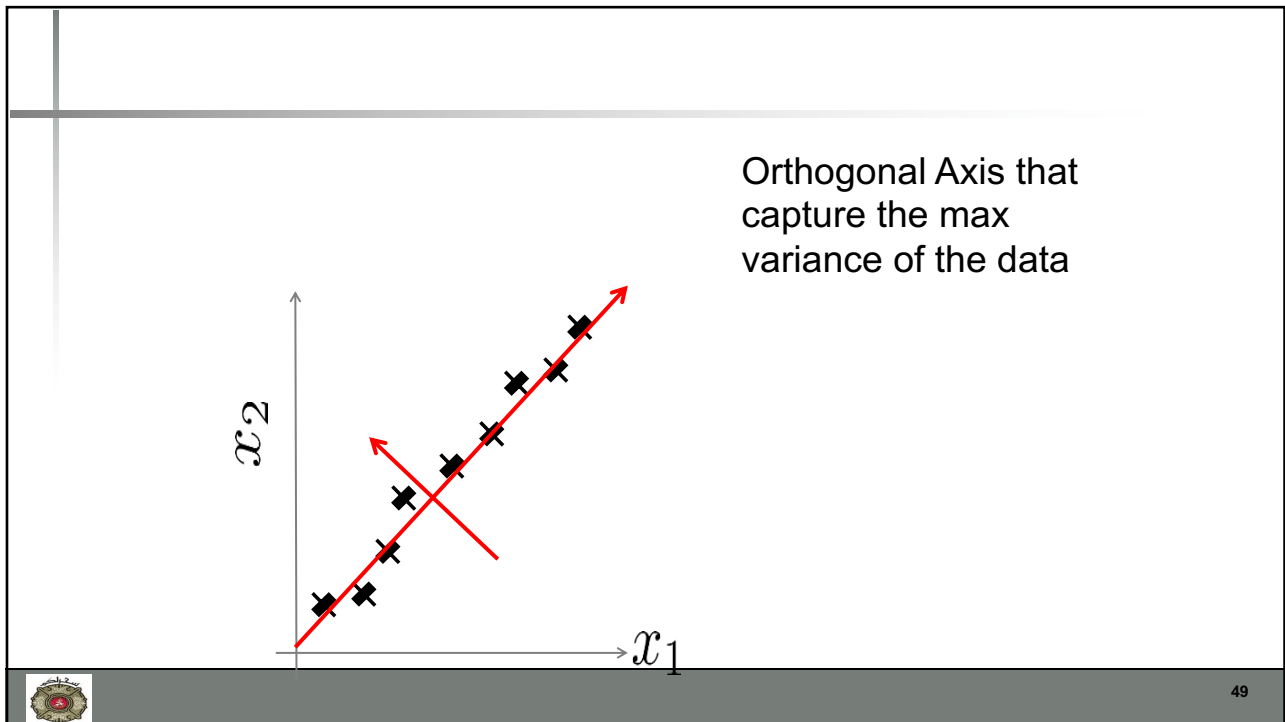
- Most common form of dimensionality reduction
- The new variables/dimensions
 - Are linear combinations of the original ones
 - Are uncorrelated with one another
- Orthogonal in original dimension space
 - Capture as much of the original **variance** in the data as possible
 - Are called Principal Components



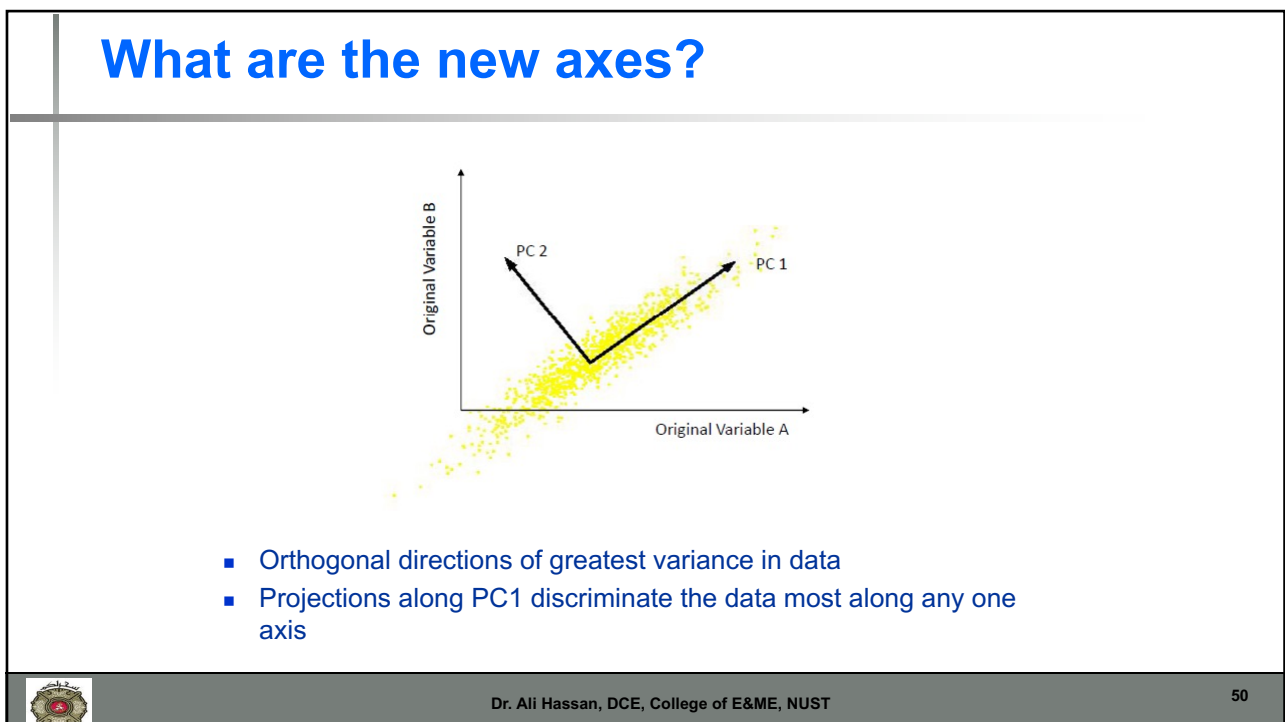
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Principal Components

- First principal component is the direction of greatest variability (covariance) in the data
- Second is the next orthogonal (uncorrelated) direction of greatest variability
 - So first remove all the variability along the first component, and then find the next direction of greatest variability
- And so on ...



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Principal Components Analysis (PCA)

- Principle
 - **Linear projection** method to reduce the number of parameters
 - Transfer a set of correlated variables into a new set of uncorrelated variables
 - Map the data into a space of lower dimensionality
 - Form of **unsupervised** learning
- Properties
 - It can be viewed as a rotation of the existing axes to new positions in the space defined by original variables
 - New axes are orthogonal and represent the directions with maximum variability



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SOME BACKGROUND OF STATISTICS



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1st order statistics

- Mean:

$$\mu_i = \frac{1}{m} \sum_{i=1}^m X_i$$

- What is the diff bw these two sets

[0 8 12 20] and [8 9 11 12]

- Std Dev:

- The average distance from the mean of the data set to a point



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1st order statistics

- Std Dev:

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (X_i - \mu)^2}{(n - 1)}}$$

- Variance (s^2)

$$\sigma^2 = \frac{\sum_{i=1}^n (X_i - \mu)^2}{(n - 1)}$$



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Covariance

- Covariance always between two dimensions $\text{cov}(x,y)$
- Covariance with itself is variance
- Covariance of 3-dimensional data set (x,y,z) ,
 - Measure cov between (x,y) , (x,z) and (y,z)



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- Variance:

$$\text{var}(X) = \frac{\sum_{i=1}^n (X_i - \mu)(X_i - \mu)}{(n - 1)}$$

- Covariance:

$$\text{cov}(X, Y) = \frac{\sum_{i=1}^n (X_i - \mu_X)(Y_i - \mu_Y)}{(n - 1)}$$



In English

- For each data item, multiply the difference between the x value and the mean of x, by the the difference between the y value and the mean of y. Add all these up, and divide by (n-1)



Question

- Is $\text{cov}(X,Y)$ equal to $\text{cov}(Y,X)$??
- $(X_i - \mu_x)(Y_i - \mu_y)$ and $(Y_i - \mu_y)(X_i - \mu_x)$ and multiplication is *commutative*



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Covariance Matrix

- For dataset with dimensions more than 2, you can calculate $\frac{n!}{(n-2)! * 2}$ different covariance values
- Calculate for $n=3$

$$C = \begin{pmatrix} \text{cov}(x, x) & \text{cov}(x, y) & \text{cov}(x, z) \\ \text{cov}(y, x) & \text{cov}(y, y) & \text{cov}(y, z) \\ \text{cov}(z, x) & \text{cov}(z, y) & \text{cov}(z, z) \end{pmatrix}$$



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Example

$$\begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 11 \\ 5 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix} \times \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 12 \\ 8 \end{pmatrix} = 4 \times \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$



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Example 2

$$2 \times \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix} \times \begin{pmatrix} 6 \\ 4 \end{pmatrix} = \begin{pmatrix} 24 \\ 16 \end{pmatrix} = 4 \times \begin{pmatrix} 6 \\ 4 \end{pmatrix}$$



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Eigen Value

- In both those examples, the amount by which the original vector was scaled after multiplication by the square matrix was the same
- Eigen value of this Eigen vector is 4



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Matrix Algebra

- Matrix * vector = rotated and scaled vector
- Matrix * vector = ONLY scaled vector and
NO rotation

Vector = Eigen vector



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Eigen Vectors

- Can only be found for **Square Matrices**
- Give $n \times n$ matrix, there are n **Eigen** vectors
- Even if we scale the Eigen vector, you get same multiple as a result—cuz you are only scaling the vector, its direction remains the same
- All Eigen vectors of a matrix are **orthogonal**
- Usually Eigen vectors are calculated as **unit vectors**: magnitude is exactly one



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Eigen Vector

$$\begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$$\sqrt{3^2 + 2^2} = \sqrt{13}$$

$$\begin{pmatrix} 3/\sqrt{13} \\ 2/\sqrt{13} \end{pmatrix}$$



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Principal Components Analysis

■ Step 1: Get some data

DATA:

<u>x</u>	<u>y</u>
2.5	2.4
0.5	0.7
2.2	2.9
1.9	2.2
3.1	3.0
2.3	2.7
2	1.6
1	1.1
1.5	1.6
1.1	0.9

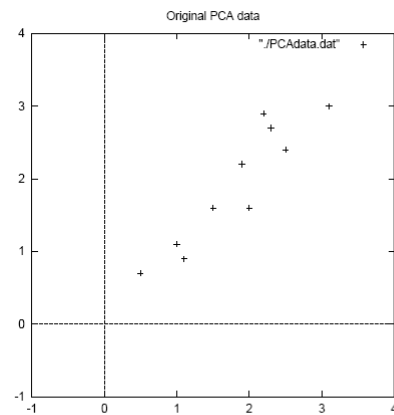


Figure 3.1: PCA example data, original data on the left, data with the means subtracted on the right, and a plot of the data



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■ Step 2

■ Subtract Mean

ZERO MEAN DATA:

<u>x</u>	<u>y</u>
.69	.49
-1.31	-1.21
.39	.99
.09	.29
1.29	1.09
.49	.79
.19	-.31
-.81	-.81
-.31	-.31
-.71	-1.01



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- Step 3

- Calculate Covariance matrix

$$\text{cov} = \begin{pmatrix} .616555556 & .615444444 \\ .615444444 & .716555556 \end{pmatrix}$$

- since the non-diagonal elements in this covariance matrix are positive, we should expect that both the x and y variable increase together.



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- Step 4:

- Calculate the eigenvectors and eigenvalues of the covariance matrix

$$\text{eigenvalues} = \begin{pmatrix} .0490833989 \\ 1.28402771 \end{pmatrix}$$

$$\text{eigenvectors} = \begin{pmatrix} -.735178656 & -.677873399 \\ .677873399 & -.735178656 \end{pmatrix}$$



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Eigen Vectors

$$U = \begin{bmatrix} | & | & & | \\ u^{(1)} & u^{(2)} & \dots & u^{(n)} \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{n \times n}$$

- Sign of the Eigen vector does not really matter
 - Vector in opposite direction
- Sort them according to eigen value



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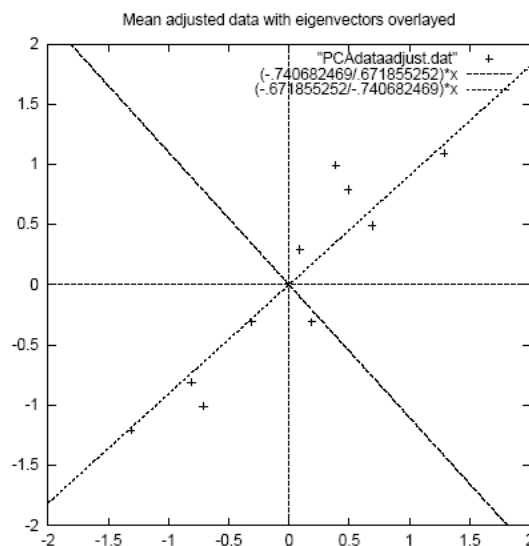


Figure 3.2: A plot of the normalised data (mean subtracted) with the eigenvectors of the covariance matrix overlayed on top.



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Note:

- Note they are perpendicular to each other.
- Note one of the eigenvectors goes through the middle of the points, like drawing a line of best fit.
- The second eigenvector gives us the other, less important, pattern in the data, that all the points follow the main line, but are off to the side of the main line by some amount.



PCA Example –STEP 5

Now, if you like, you can decide to *ignore* the components of lesser significance.

You do lose some information, but if the eigenvalues are small, you don't lose much

- n dimensions in your data
- calculate n eigenvectors and eigenvalues
- choose only the first k eigenvectors
- final data set has only k dimensions.



PCA Example –STEP 5

- Feature Vector

FeatureVector = (eig₁ eig₂ eig₃ ... eig_n)

We can either form a feature vector with both of the eigenvectors:

$$\begin{pmatrix} -.677873399 & -.735178656 \\ -.735178656 & .677873399 \end{pmatrix}$$

or, we can choose to leave out the smaller, less significant component and only have a single column:

$$\begin{pmatrix} -.677873399 \\ -.735178656 \end{pmatrix}$$



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Eigen Vectors

$$U = \begin{bmatrix} | & | & & | \\ u^{(1)} & u^{(2)} & \dots & u^{(n)} \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{n \times n}$$

- Ureduce = U(:,1:k) ;

- z = Ureduce' * x; % Projected data



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PCA Example –STEP 5

- Deriving the new data

FinalData = RowFeatureVector x RowZeroMeanData

RowFeatureVector is the matrix with the eigenvectors in the columns *transposed* so that the eigenvectors are now in the rows, with the most significant eigenvector at the top

RowZeroMeanData is the mean-adjusted data *transposed*, ie. the data items are in each column, with each row holding a separate dimension.



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PCA Example –STEP 5

FinalData transpose: dimensions along columns

x	y
-.827970186	-.175115307
1.77758033	.142857227
-.992197494	.384374989
-.274210416	.130417207
-1.67580142	-.209498461
-.912949103	.175282444
.0991094375	-.349824698
1.14457216	.0464172582
.438046137	.0177646297
1.22382056	-.162675287



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PCA Example –STEP 5

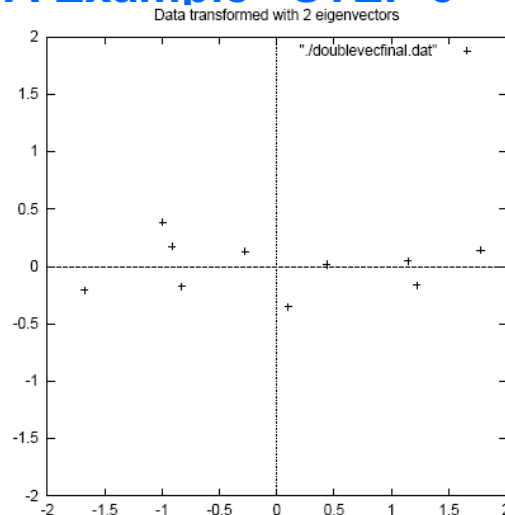


Figure 3.3: The table of data by applying the PCA analysis using both eigenvectors, and a plot of the new data points.

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Reconstruction of original Data

- If we reduced the dimensionality, obviously, when reconstructing the data we would lose those dimensions we chose to discard. In our example let us assume that we considered only the x dimension...



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- $Z = U^T * X$
- $X' = (U^{-1} * Z) + \text{originalMean}$



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Reconstruction of original Data

X

-0.827970186
 1.77758033
 -0.992197494
 -0.274210416
 -1.67580142
 -0.912949103
 0.0991094375
 1.14457216
 0.438046137
 1.22382056

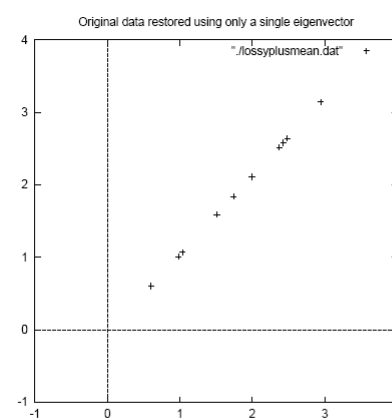


Figure 3.5: The reconstruction from the data that was derived using only a single eigenvector



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HOW TO SELECT K PCS



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PCs, Variance and Least-Squares

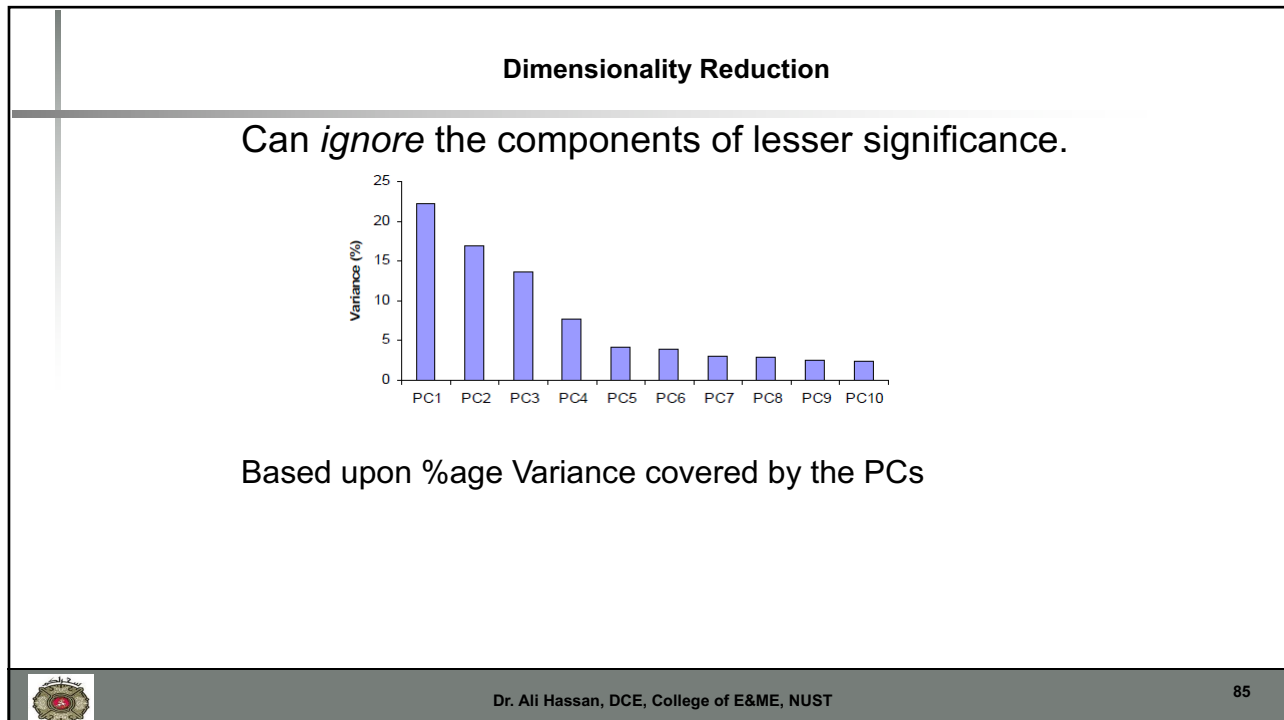
- The first PC retains the greatest amount of variation in the sample
- The k th PC retains the k th greatest fraction of the variation in the sample
- The k th largest eigenvalue of the correlation matrix C is the variance in the sample along the k th PC
- The least-squares view: PCs are a series of linear least squares fits to a sample, each orthogonal to all previous ones



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$$S = \begin{bmatrix} s_{11} & 0 & 0 & 0 & 0 \\ 0 & s_{22} & 0 & 0 & 0 \\ 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & s_{nn} \end{bmatrix}$$

$$\frac{\sum_{i=1}^k s_{ii}}{\sum_{i=1}^n s_{ii}} \geq 0.99$$

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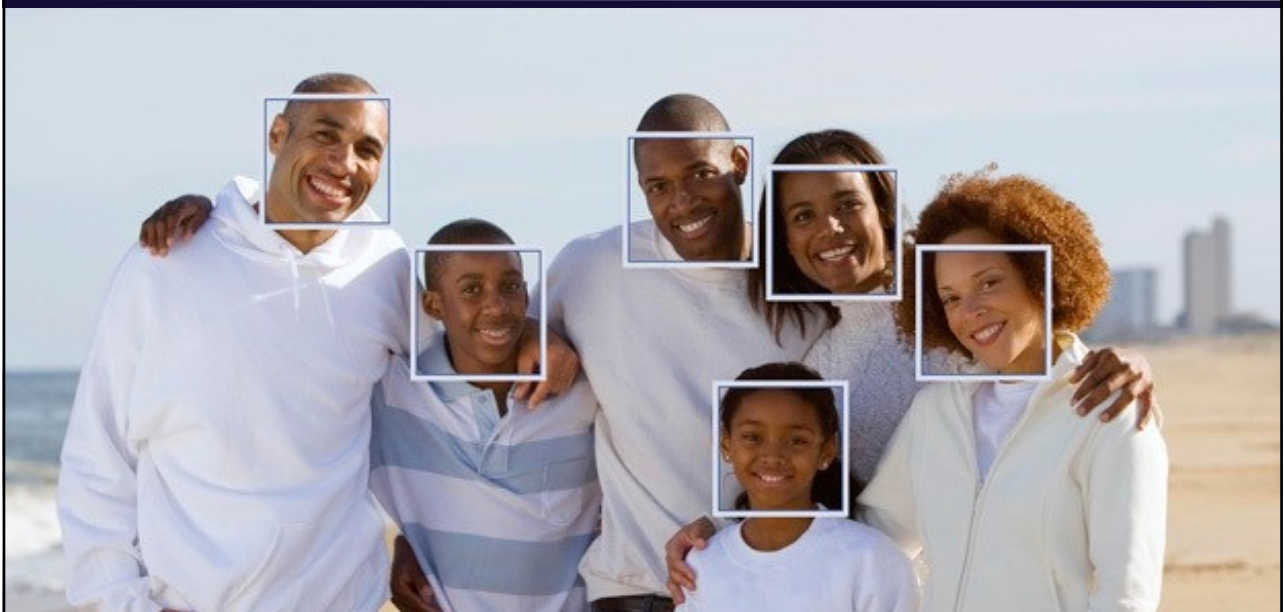
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Facial Recognition using Eigen Vectors

- **Face Detection**
- **Face Recognition**
 - Face Verification
 - Facial Identification or recognition

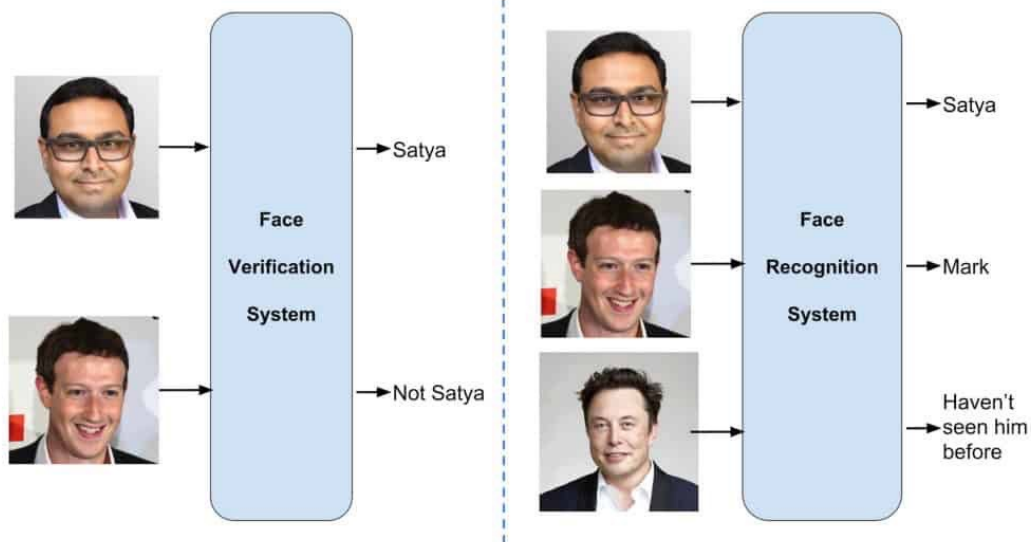
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Face Detection



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Face Recognition & Verification



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Eigen Faces

Training images

$x_1, \dots, x_N$

Face Recognition Using Eigenfaces
Matthew A. Turk and Alex P. Pentland
Vision and Modeling Group, The Media Laboratory
Massachusetts Institute of Technology

<https://sites.cs.ucsb.edu/~mturk/Papers/mturk-CVPR91.pdf>



Bill Gates	Arnold Schwarzenegger	Gwyneth Paltrow	Angelina Jolie
Queen Elizabeth II	David Beckham	LeBron James	Michael Jordan
Michael Jackson	Dwayne Johnson	Marilyn Monroe	Azra Akin
Hillary Clinton	Oprah Winfrey	George W Bush	Daniel Radcliffe

dataset of 1000 faces

https://pvimagesearch.com/wp-content/uploads/2021/05/kirby_1987.pdf

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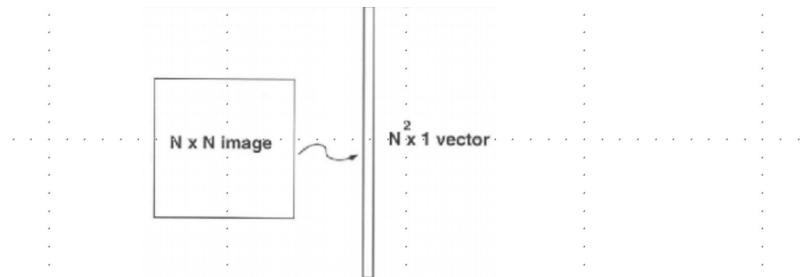
Mean Face

Mean: μ



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Algorithm



• Main idea behind eigenfaces

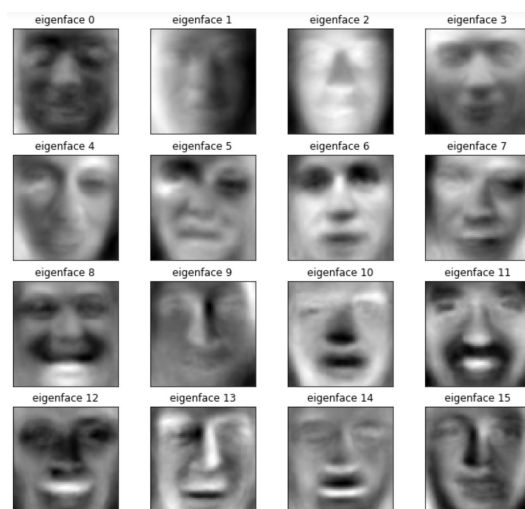
- Suppose Γ is an $N^2 \times 1$ vector, corresponding to an $N \times N$ face image I .
- The idea is to represent Γ ($\Phi = \Gamma - \text{mean face}$) into a low-dimensional space:

$$\hat{\Phi} - \text{mean} = w_1 u_1 + w_2 u_2 + \dots + w_K u_K \quad (K \ll N^2)$$

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Eigen Faces

Top eigenvectors: $u_1, \dots, u_k$



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Reconstruction



$$\mathbf{x} \rightarrow [\mathbf{u}_1^T(\mathbf{x} - \mu), \dots, \mathbf{u}_k^T(\mathbf{x} - \mu)]$$

$$= w_1, \dots, w_k$$

This vector is the representation of the face.

Top eigenvectors: $\mathbf{u}_1, \dots, \mathbf{u}_k$

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Reconstruction

- Face $\mathbf{x}$ in “face space” coordinates:



$$\mathbf{x} \rightarrow [\mathbf{u}_1^T(\mathbf{x} - \mu), \dots, \mathbf{u}_k^T(\mathbf{x} - \mu)]$$

$$= w_1, \dots, w_k$$

This vector is the representation of the face.

- Reconstruction:



$$\hat{\mathbf{x}} = \mu + w_1\mathbf{u}_1 + w_2\mathbf{u}_2 + w_3\mathbf{u}_3 + w_4\mathbf{u}_4 + \dots$$

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Reconstruction with 50 Eigen Faces



1000 x 64 x 64 to 50 x 64 x 64

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Reconstruction from 100 Eigen Faces



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Reconstruction with 250 Eigen Faces

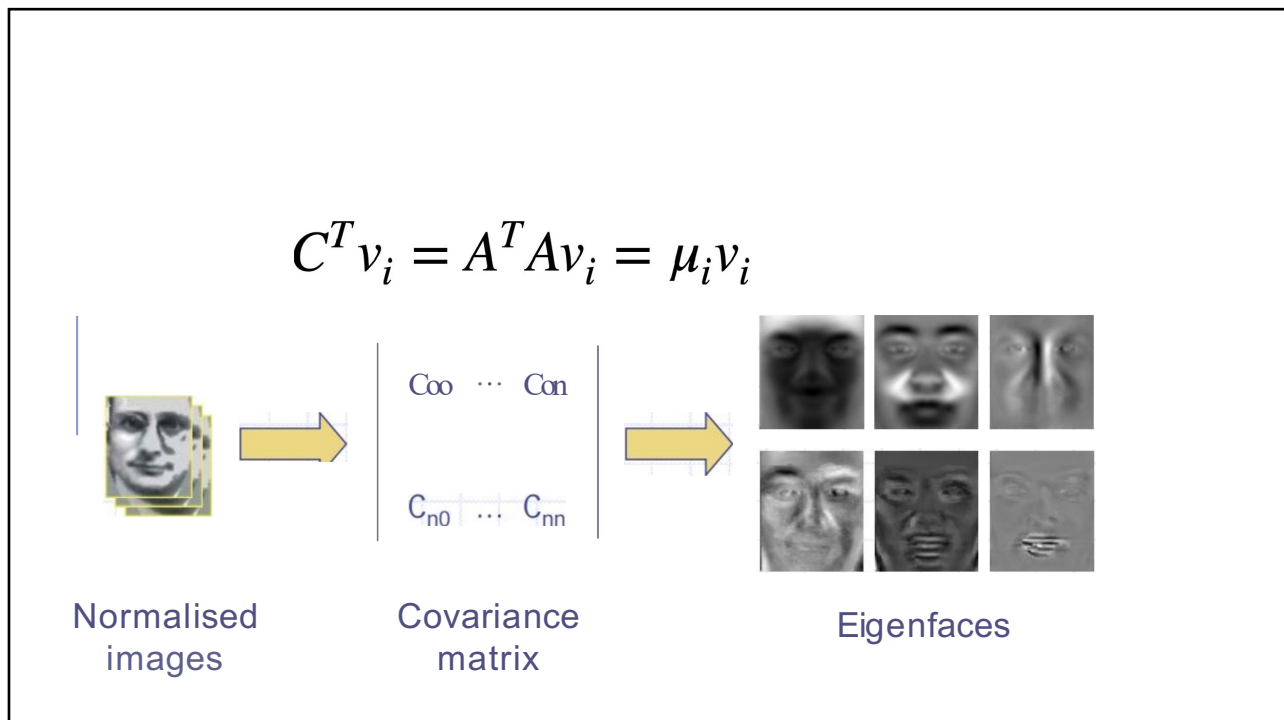


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Reconstruction from 1000 Eigen Faces



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The dimensionality trick

So $C = AA^T$ is a huge matrix.

But consider $A^T A$. It is only $M \times M$. Finding those eigenvalues is easy.

Suppose v_i is an eigenvector $A^T A$:

$$A^T A v_i = \lambda v_i$$

Premultiply by A :

$$AA^T A v_i = \lambda A v_i$$

So: $A v_i$ are the eigenvectors of $C = AA^T$

$N^2 \times K$ $K \times 1$

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chatGPT Prompt

Generate python code that takes 8 samples of face images, find eigen faces and generate the faces again from first four eigen faces

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Python Code

```
import numpy as np
import cv2
from sklearn.decomposition import PCA
from sklearn.datasets import fetch_lfw_people
import matplotlib.pyplot as plt

# Load face images (You can replace this with your own dataset)
lfw_people = fetch_lfw_people(min_faces_per_person=8, resize=0.4)
faces = lfw_people.images

# Reshape the face images into a 2D array
n_samples, h, w = faces.shape
X = faces.reshape(n_samples, -1)

# Number of eigenfaces to keep
n_components = 4

# Perform PCA to find the eigenfaces
pca = PCA(n_components=n_components, svd_solver='randomized', whiten=True).fit(X)

# Get the eigenfaces
eigenfaces = pca.components_.reshape((n_components, h, w))

# Project the data onto the lower-dimensional space defined by the principal components
X_pca = pca.transform(X)

# Reconstruct the faces using the first four eigenfaces
reconstructed_faces = pca.inverse_transform(X_pca)

# Reshape the reconstructed faces to the original shape
reconstructed_faces = reconstructed_faces.reshape(n_samples, h, w)

# Project the data onto the lower-dimensional space defined by the principal components
X_pca = pca.transform(X)

# Reconstruct the faces using the first four eigenfaces
reconstructed_faces = pca.inverse_transform(X_pca)

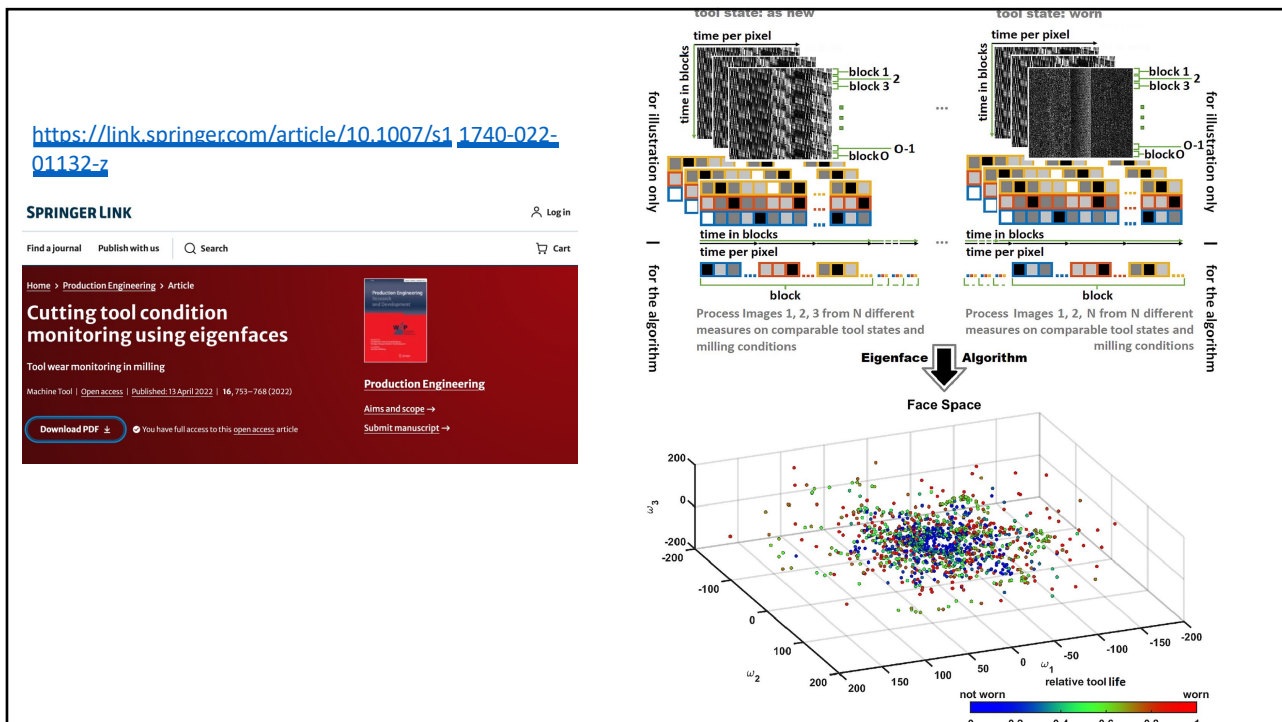
# Reshape the reconstructed faces to the original shape
reconstructed_faces = reconstructed_faces.reshape(n_samples, h, w)

# Plot the original and reconstructed faces
plt.figure(figsize=(8, 4))
for i in range(8):
    plt.subplot(8, 2, 2 * i + 1)
    plt.imshow(faces[i], cmap=plt.cm.bone)
    plt.title("Original Face")
    plt.xticks(())
    plt.yticks(())

    plt.subplot(8, 2, 2 * i + 2)
    plt.imshow(reconstructed_faces[i], cmap=plt.cm.bone)
    plt.title("Reconstructed Face")
    plt.xticks(())
    plt.yticks(())

plt.tight_layout()
plt.show()
```

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Reading Material

- **Chapter 5: Feature Selection Methods and Results** by Ali Hassan
- **Parcel: Feature Subset Selection in Variable Cost Domains**, Cambridge University Technical Report by M. Scott, M. Niranjan
- **A tutorial on Principal Components Analysis** by Lindsay I Smith



END OF CLASS

